Performance Evaluation of YOLOv10 for Prohibited Item Detection in X-ray Imagery

Aryam E. Mersha - FTP0220/14
Addis Ababa Science and Technology University
Submitted to: Dr. Ashenafi G.

***Abstract*—This paper presents a practical evaluation of the recently proposed YOLOv10, a state-of-the-art, end-to-end object detector. We investigate its practical application for the specialized task of identifying prohibited items in X-ray security scans. This work assesses two model variants, YOLOv10-m (medium) and YOLOv10-l (large), on the comprehensive Few-Shot Object Detection (FSOD) dataset. A key innovation of YOLOv10 is its NMS-free training strategy, which we test by training models for 60 epochs and evaluating them on standard metrics including Precision, Recall, F1-Score, and mean Average Precision (mAP). Our results validate the effectiveness of the architecture, with the YOLOv10-l model achieving a superior mAP@50-95 of 0.559 on the held-out test set, providing a strong baseline for the use of end-to-end YOLO models in security contexts.**

***Index Terms*—Object Detection, YOLOv10, NMS-Free, X-ray Security Screening, Deep Learning, FSOD Dataset.**

I. INTRODUCTION

Object detection is a fundamental computer vision task that involves identifying and localizing multiple objects within an image. Its applications are widespread, from autonomous driving to smart healthcare [1]. The "You Only Look Once" (YOLO) series, first proposed by Joseph Redmon, revolutionized the field by reframing detection as a single regression problem, enabling real-time performance [1]. Each subsequent version has improved the critical balance between speed and accuracy.

YOLOv10, introduced by Wang et al. [2], marks a significant leap by creating a truly end-to-end detector. It addresses a key performance bottleneck in previous models: the reliance on Non-Maximum Suppression (NMS) for post-processing. This study provides a practical validation of YOLOv10's architectural innovations by applying it to the challenging domain of X-ray security screening.

II. YOLOv10 ARCHITECTURE AND INNOVATIONS

YOLOv10 achieves its state-of-the-art performance through two primary advancements: a novel training strategy for NMS-free inference and a holistically optimized model architecture [2].

A. NMS-Free Training with Consistent Dual Assignments

A major latency bottleneck in previous YOLO models was the NMS step. YOLOv10 eliminates this by employing a dual-assignment training strategy with two heads:

1) **One-to-Many Head:** Similar to traditional YOLOs, this provides rich, supervisory signals during training, facilitating robust feature learning.
2) **One-to-One Head:** This ensures each ground-truth object corresponds to a single positive prediction. During inference, only this head is used, making the NMS step entirely unnecessary and significantly reducing latency [2].

B. Holistic Efficiency-Accuracy Driven Design

YOLOv10 meticulously re-evaluates and optimizes its components for both speed and accuracy.

- **Lightweight Classification Head:** Recognizing that regression is often more critical for performance, YOLOv10 uses a simpler, lightweight architecture for the classification head, reducing computational overhead.
- **Spatial-Channel Decoupled Downsampling (SCDown):** This replaces standard convolutions with a more efficient method that separates channel transformation and spatial reduction, preserving information while reducing parameters [2].
- **Partial Self-Attention (PSA):** To efficiently model global context, PSA is integrated after the main backbone. It partitions features and applies multi-head self-attention to only one part, significantly reducing the computational cost typically associated with attention mechanisms [2].

III. EXPERIMENTAL SETUP & RESULTS

A. Dataset

The study utilizes the Few-Shot Object Detection (FSOD) dataset, which contains X-ray images of luggage with 20 classes of prohibited or notable items. The dataset is pre-divided into training, validation (val), and a held-out test set (newtest), enabling a robust evaluation of model generalization.

B. Training and Evaluation

Both the YOLOv10-m and YOLOv10-l models were trained for 60 epochs on a Kaggle platform using an NVIDIA Tesla T4 GPU. Performance was evaluated using Precision (P), Recall (R), F1-Score, mAP@50, and mAP@50-95.

C. Performance Analysis

The quantitative results for both models on the validation and newtest sets are presented in Table I. The YOLOv10-l model demonstrates superior performance across all primary metrics, confirming that its larger capacity translates to higher accuracy. The consistent performance between the validation and newtest sets indicates that both models generalize well and do not suffer from significant overfitting.

Visual analysis of the final model performance on the unseen ‘newtest‘ set provides further insight. The Precision-Recall (PR) curves in Fig. 1 illustrate the trade-off between the metrics, with YOLOv10-l generally maintaining higher precision at various recall levels. The confusion matrices in Fig. 2 identify specific class-level successes and failures, such as the models’ strong ability to detect ‘laptops‘ versus the relative difficulty with ‘folding_knife‘.

IV. CONCLUSION

This study successfully validated the performance of the end-to-end YOLOv10 detector on the challenging FSOD X-ray dataset. The empirical results confirm the effectiveness of YOLOv10’s architectural innovations, with both the medium and large variants achieving high accuracy. The YOLOv10-l model, in particular, establishes a powerful performance baseline for automated threat detection. The successful application of this NMS-free model demonstrates a significant step forward in building faster and more efficient object detection pipelines for real-world security applications.

REFERENCES

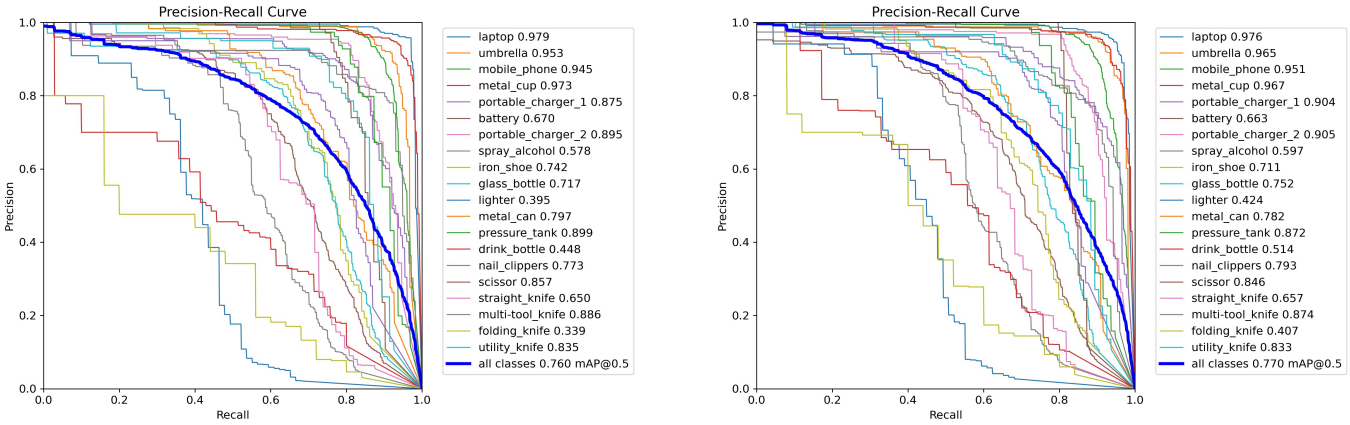
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TABLE I: PERFORMANCE METRICS ON FSOD DATASET

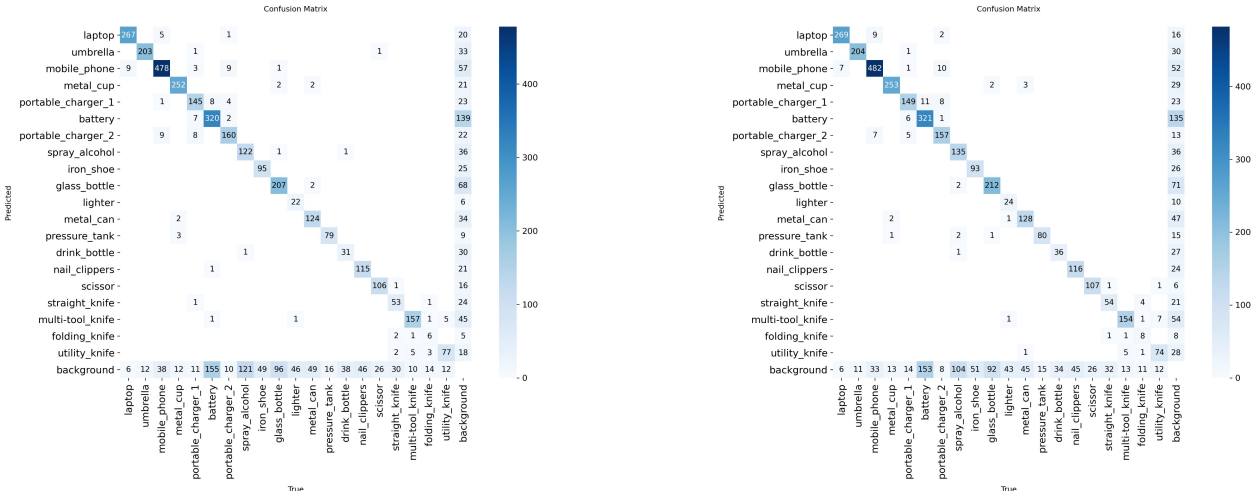
Model	Dataset	Precision	Recall	F1-Score	mAP@50	mAP@.5-.95
YOLOv10-m	Validation	0.817	0.680	0.742	0.762	0.541
	New Test	0.795	0.703	0.746	0.760	0.541
YOLOv10-l	Validation	0.807	0.711	0.756	0.778	0.559
	New Test	0.807	0.711	0.756	0.778	0.559



(a) YOLOv10-m on newtest set

(b) YOLOv10-l on newtest set

Fig. 1: Precision-Recall Curve for (a) YOLOv10-m and (b) YOLOv10-l on the newtest set.



(a) YOLOv10-m on newtest set

(b) YOLOv10-l on newtest set

Fig. 2: Confusion Matrix for (a) YOLOv10-m and (b) YOLOv10-l on the newtest set.